

PLC CONTROLLED STAR DELTA MOTOR SYSTEM
WITH AUTOMATIC TRANSFER SWITCH AND PID SYSTEM

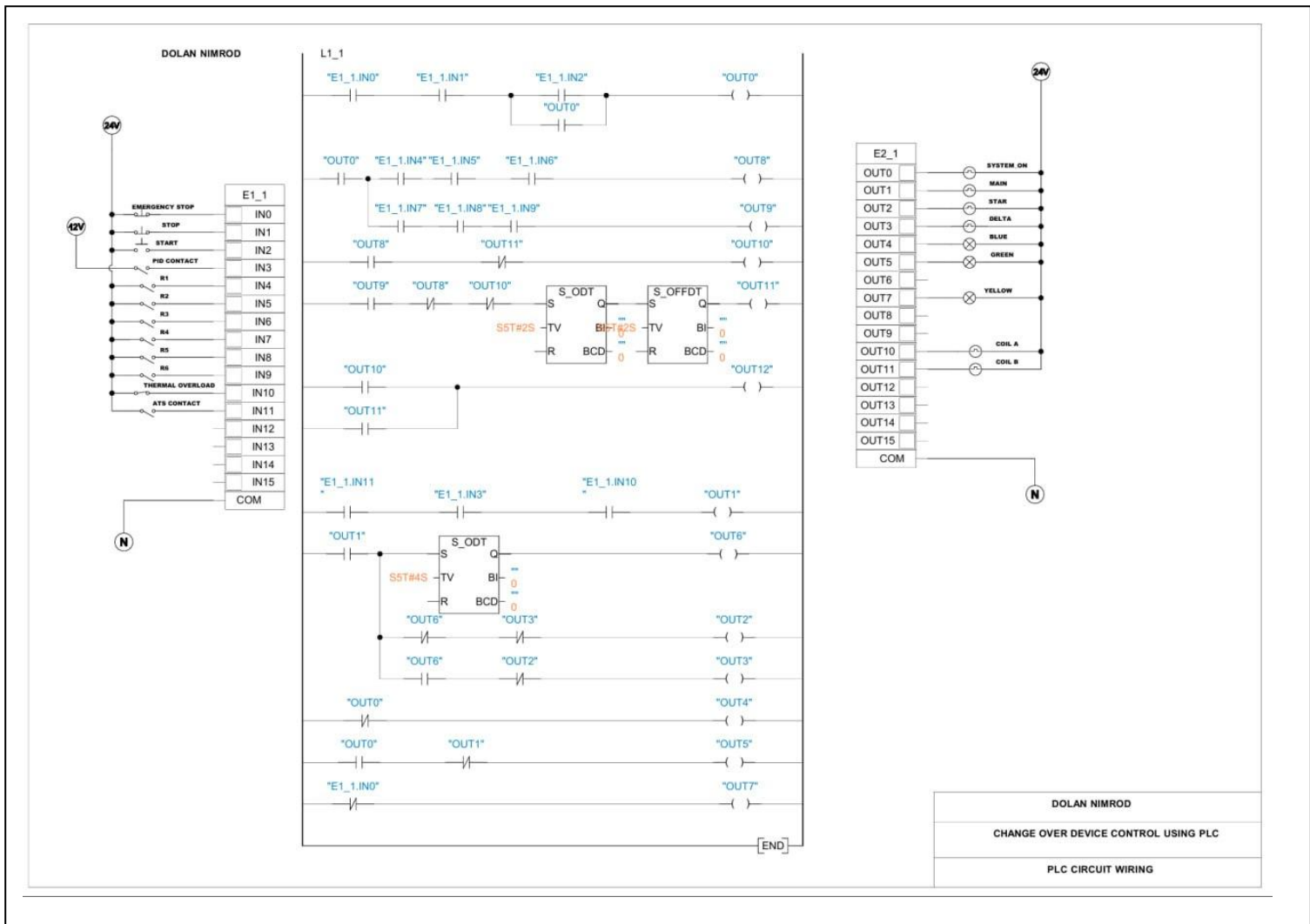
Objective

Design a system that automatically switches between a main power supply and a backup power supply using an Automatic Transfer Switch (ATS)- a changeover device to control a three-phase motor using a star-delta starting method.

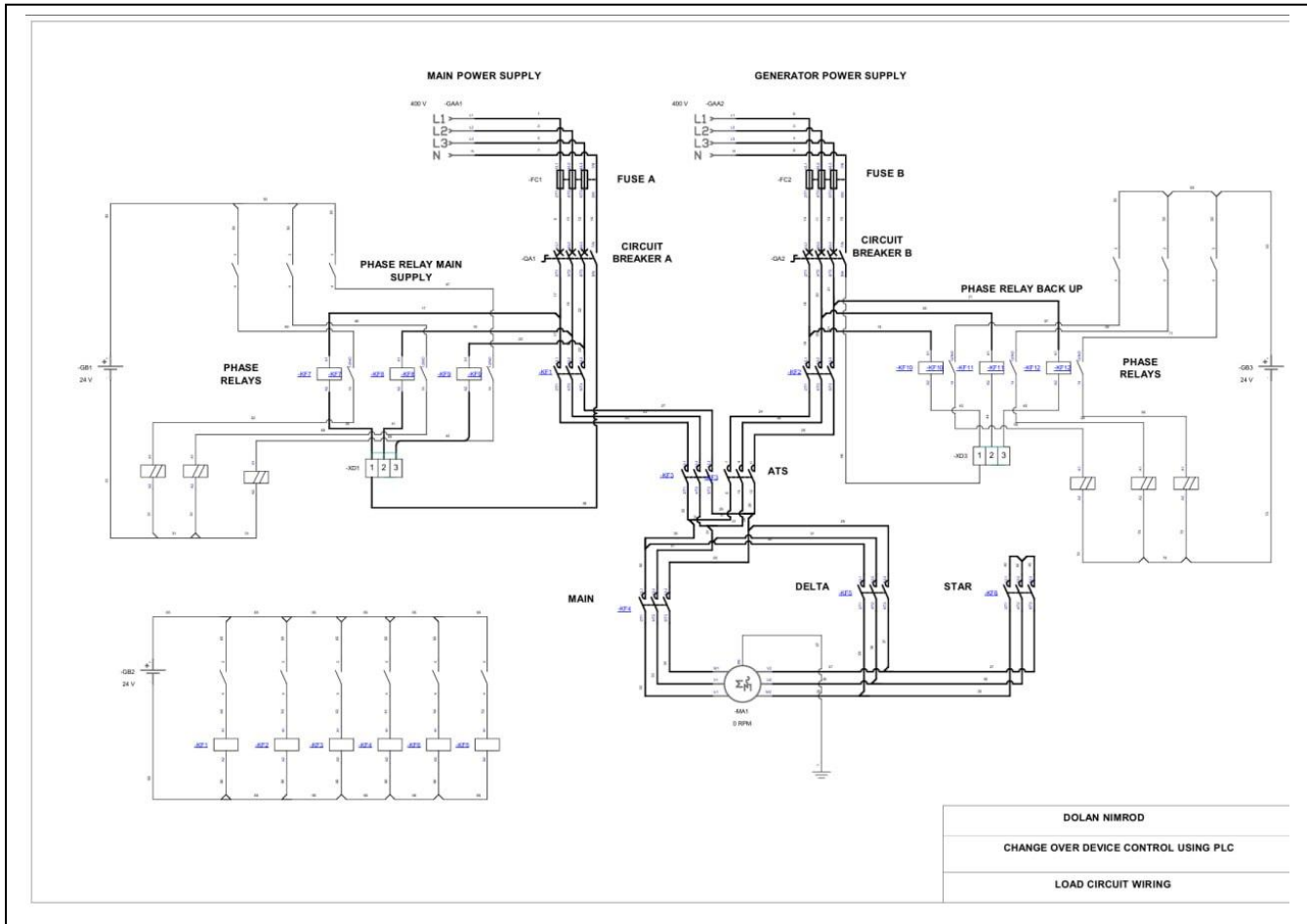
Components

PLC Controller	1
PLC power supply	1
Relay Modules	6
PID Temperature Controller	1
Contactors	5
Push buttons	3
Thermal Overload Relay	1
Change Over Device	1
Indicator Lamps	5

CONTROL DIAGRAM



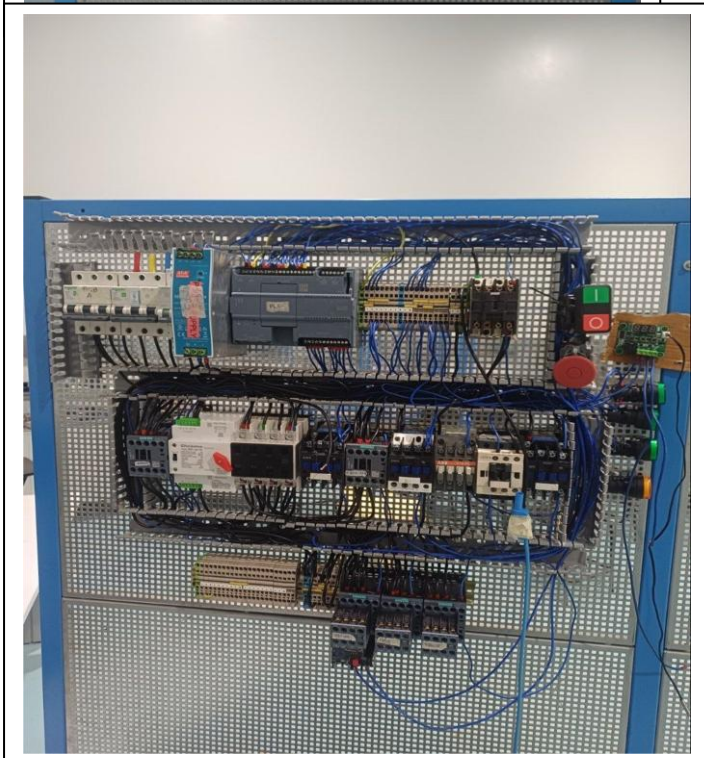
LOAD CIRCUIT DIAGRAM



SET UP RACK



Rack set up



Wiring of electrical components before a whole modifications.

Challenges encountered

1. Proper component design placement
2. 240AC contactor faults
3. Three-phase supply wiring from the source.
4. Control switch panel board.

Recommendation

1. Test all components before use to ensure they are functioning properly.
2. Measure the amount of voltage being supplied from each phase to avoid any faulty line voltage.
3. Ensure components are well placed for a clear ordered workplace.
4. Draw a well circuit designed diagram before any installations.